AI in Business: Analytics, Ethics & Adoption

Complete answers in easy language with examples. Each answer starts with a simple definition, then key points, then an example you can write in the exam. **Bold blue words** are keywords – examiners look for them!

No.	Question (topic)	Type
1 – 9	Descriptive, predictive, prescriptive analytics, KPI, AI bias, responsible AI, data privacy, generative AI, data governance	Section A – short
10 – 12	Three types of analytics with industries, AI & business strategy, KPIs for AI impact	Section B – medium
13 – 16	AI ethics & trust, AI adoption challenges & strategies, generative AI & automation, AI & customer satisfaction	Section C – long
17 – 19	Real-life case study (UPS ORION), data protection & governance, legal / ethical / social implications	Section D – case & essays

Duplicate questions – merged (so you study once!)

- "Discuss the challenges businesses face when adopting AI" (asked 2 times) + "Key risks and challenges... strategies for data, talent, cost, stakeholder resistance" + "Risks and challenges during AI adoption" = **one answer: Q14**
- "What is Generative AI?" (asked 2 times) = **Q8** (short) and **Q15** (long)
- "Differentiate descriptive, predictive, prescriptive with two industries" + "Explain the three types of business analytics... how do they support AI decision-making" = **one answer: Q10**
- "Data governance" (short) = **Q9**; "data protection & data governance" (long) = **Q18**

How to write a good answer (scoring tips)

- ✅ **2–3 marks:** definition + 1 example (4–6 lines)
- ✅ **5 marks:** definition + 4–5 points + example or small diagram
- ✅ **10 marks:** introduction, headings, 6–8 points, table / diagram, real company example, conclusion
- ✅ Underline keywords, draw neat diagrams, and always add an **Indian or global company example**

Section A – Short Answer Questions

Q1 Define descriptive analytics and give one example of its business application.

Descriptive analytics means studying past data to understand "What happened?" It summarises data using totals, averages, percentages, charts, reports and dashboards. It does not predict the future – it only describes what already happened.

- Tools: Excel, Tableau / Power BI dashboards, SQL reports
- Techniques: averages, growth %, charts, data aggregation

Example

A retail chain's ***monthly sales dashboard*** shows that sales fell 10% in March compared to February, and that the Ludhiana branch and the winter-wear category fell the most. The manager now knows WHAT happened and WHERE.

Q2 What is predictive analytics? State one use case in the retail industry.

Predictive analytics uses historical data, statistics and machine learning to forecast "What is likely to happen?" It finds patterns in old data and uses them to estimate future outcomes with a probability (e.g. 80% chance).

- Techniques: regression, time-series forecasting, classification, machine learning models

Retail use case - Demand forecasting

A supermarket uses past sales, weather, prices and the festival calendar (Diwali, Eid) to ***predict how many units of each product every store will sell next month***. This avoids stock-outs (empty shelves) and over-stock (waste). Other retail uses: predicting which customers may stop buying (***churn prediction***).

Q3 What is prescriptive analytics? How does it differ from predictive analytics?

Prescriptive analytics goes one step further and answers "What should we do?" It recommends the ***best action*** using optimisation, simulation and AI, often after testing many options ("what-if").

Basis	Predictive analytics	Prescriptive analytics
Question	What WILL happen?	What SHOULD we do?
Output	A forecast / probability	A recommended action or decision
Techniques	Regression, ML, time-series	Optimisation, simulation, rules, reinforcement learning
Example	"Next week demand =500 units"	"Order 520 units from Supplier B by Tuesday - lowest cost"
Complexity	Medium	High (uses predictions as input)

Q4 Define KPI (Key Performance Indicator) and name two examples used to measure AI impact.

A ***KPI*** is a measurable value that shows how well a person, team or company is achieving an important goal. Good KPIs are ***SMART*** - Specific, Measurable, Achievable, Relevant, Time-bound.

- ***Cost savings / ROI from AI*** - e.g. Rs 30 lakh saved per year after automating invoice processing
- ***Customer satisfaction (CSAT / NPS)*** - e.g. CSAT rises from 72% to 85% after an AI chatbot
- Other KPIs: forecast accuracy, process time saved, chatbot resolution rate, machine downtime

Q5 What is meant by 'AI bias'? Give one real-world example.

AI bias means an AI system gives unfair or one-sided results that favour or harm certain groups (gender, caste, religion, race, age, region). It usually happens because the **training data** carries old human bias, or the data is incomplete. "Biased data in = biased decisions out."

Real-world example - Amazon hiring tool

Amazon built an experimental AI tool to screen job resumes. It learned from 10 years of past resumes, which came mostly from men. The tool started giving lower scores to resumes containing the word "women's" (e.g. "women's chess club"). Amazon stopped using it (reported by Reuters in 2018).

Q6 Define 'responsible AI' and state two principles it is based on.

Responsible AI means designing, building and using AI in a way that is safe, fair, transparent and accountable, respects people's rights and follows the law. In India, NITI Aayog promotes this idea as "Responsible AI for All".

Fairness | Transparency | Accountability | Privacy & Security | Reliability & Safety | Inclusive-ness

- **Fairness** - AI must not discriminate; test it for bias regularly
- **Transparency / Explainability** - people should know when AI is used and why it gave a decision
- **Accountability** - a human or team is responsible for AI outcomes

Q7 What is data privacy? Why is it a concern in AI-driven business systems?

Data privacy is the right of a person to control how their personal data (name, phone, location, health, purchases) is collected, used, stored and shared - only with **consent** and for a clear purpose.

Why is it a concern in AI?

- AI needs **huge amounts of personal data** - more data collected = more risk
- AI can **guess sensitive facts** (health, income, religion) from simple data like shopping history
- Risk of **data breaches / hacking** and data being sold or shared without consent
- Laws like India's **DPDP Act 2023** and Europe's **GDPR** give heavy penalties
- If customers feel watched, they **lose trust** and leave

Q8 What is generative AI? Name one popular generative AI tool. (also: "What is Generative AI?")

Generative AI is a type of AI that creates new content - text, images, music, video or computer code - after learning patterns from very large amounts of data. You give it an instruction called a **prompt**, and it generates an answer.

Prompt ("Write an ad for our new shoes") → **GenAI model** (Large Language Model (LLM)) → **New content** (Ad copy, image, code, video)

Popular tools: ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), Microsoft Copilot; images – Midjourney, DALL-E. Limitation: it can produce wrong facts confidently ("hallucination"), so a human must check the output.

- **Marketing:** ad copy, social media posts, product images in minutes
- **Customer support:** chatbots that answer in natural language
- **Office work:** summarising long reports, writing emails, making slides
- **Software:** writing and fixing code; **Design:** logos, product mock-ups

Q9 What is data governance? State its importance in AI implementation.

Data governance is the set of rules, roles, policies and processes that make sure an organisation's data is **accurate, secure, consistent and used properly** – from collection to deletion. It answers: Who owns the data? Who can use it? Is it correct? Is it legal?

Importance in AI implementation

- ✅ **Good data = good AI** – "garbage in, garbage out"; clean data gives accurate models
- ✅ **Legal compliance** with privacy laws (DPDP Act, GDPR)
- ✅ **Security** – only authorised people access sensitive data
- ✅ **Accountability & trust** – every dataset has an owner; decisions can be audited and explained

Example – a bank

The bank names a **data owner** for customer data, a **data steward** who checks quality (no duplicate accounts, correct addresses) and an IT **data custodian** who stores it securely. Aadhaar and PAN numbers are masked before data is used to train a loan-approval AI.

Section B – Medium Answer Questions

Q10 Differentiate between descriptive, predictive and prescriptive analytics with examples from at least two industries. How do they support AI-driven decision-making in an organisation? (merged question)

Business analytics is the use of data, statistics and technology to make better business decisions. It has three main types. They form a ladder – each step gives more value but needs more data and skill.



Basis	Descriptive	Predictive	Prescriptive
Meaning	Summarises past data	Forecasts future outcomes	Recommends best action

Basis	Descriptive	Predictive	Prescriptive
Data used	Historical	Historical + current	Historical + predictions + rules
Tools	Excel, Tableau, SQL	Regression, ML, Python	Optimisation, simulation, AI
Retail	Last Diwali sales by category & store	Forecast this Diwali's demand	How much stock to send to each store & what discount to give
Healthcare	Patients admitted per month, bed occupancy	Which patients are at high risk of re-admission	Suggest follow-up calls, treatment plan & nurse schedule
Banking	Number of loan defaults last year	Probability that a new applicant will default	Approve / reject the loan and set the interest rate

Extra marks: the 4th type

Diagnostic analytics sits between descriptive and predictive. It answers ***"Why did it happen?"*** using drill-down and root-cause analysis – e.g. sales fell because a competitor opened nearby and our prices were higher.

How they support AI-driven decision-making

1. ***Descriptive builds the foundation*** – clean data, KPIs and dashboards show the current situation. AI cannot work without this base.
2. ***Predictive gives foresight*** – machine learning models warn managers early (falling demand, fraud, machine failure), so they act before the problem happens.
3. ***Prescriptive turns insight into action*** – AI recommends or even automatically takes the best decision (price, route, stock order).
4. ***Together they form a loop*** – action creates new data, which is described, predicted and optimised again, so decisions keep improving.
5. Result: decisions become ***faster, data-driven and consistent***, and less dependent on gut feeling.

Example – Airline pricing

Descriptive: past seat bookings per route. Predictive: expected demand for the Delhi–Mumbai flight next Friday. Prescriptive: AI sets the ticket price every hour (***dynamic pricing***) to fill seats and maximise revenue.

Q11 ***How does AI-driven decision-making improve business strategy? Illustrate with a case example.***

AI-driven decision-making means using AI and data analysis to support or automate business decisions, instead of depending only on experience or gut feeling.



How it improves strategy

- **Better accuracy** – decisions based on evidence, not guesses
- **Speed** – AI analyses millions of records in seconds; real-time decisions
- **Better forecasting** – helps plan budgets, production, hiring and expansion
- **Personalisation** – understand each customer; a strong competitive advantage
- **Risk management** – early warning of fraud, defaults, supply problems
- **Resource optimisation** – right stock, right staff, lowest cost
- **What-if analysis** – test strategies (new price, new market) before spending money

Case example – Netflix

- **Problem:** keep subscribers watching and decide which shows to invest in.
- **AI use:** analyses viewing history, time of day, pauses and ratings to recommend shows to each user. Netflix researchers reported that about 80% of the hours people stream come from these recommendations.
- **Strategic use:** the same data guided its decision to commission "House of Cards" (2013) – data showed a strong audience for the director, the lead actor and the original British series.
- **Result:** higher engagement, fewer cancellations (lower churn) and smarter content investment.

Second example – Amazon (short)

- **Dynamic pricing:** AI checks demand, stock and competitor prices and updates prices many times a day
- **Inventory placement:** demand forecasts decide which products to keep in which warehouse, so delivery is faster
- **Strategy impact:** lower costs + faster delivery + personalised shopping = customer loyalty

Conclusion: AI moves strategy from "we think" to "the data shows". It gives speed, accuracy and personalisation, but managers must still set goals, check AI results and take final responsibility.

Q12 ***Explain how organisations use KPIs to measure the business impact of AI adoption. Give examples from manufacturing and retail.***

After investing in AI, a company must prove that AI is actually creating value. **KPIs** convert this into numbers that managers can track. Steps:

1. **Set a clear goal** – e.g. "reduce machine breakdowns" (linked to business strategy)
2. **Measure the baseline** – record the KPI value BEFORE AI
3. **Choose the right KPIs** – financial, operational, customer, employee and model KPIs
4. **Compare after AI** – pilot vs non-pilot, or A/B testing
5. **Track on dashboards** and review every month / quarter
6. **Calculate ROI** = (Benefit from AI – Cost of AI) / Cost of AI x 100

Quick ROI example

AI system cost = Rs 20 lakh. Savings in one year = Rs 50 lakh. ROI = (50 – 20) / 20 x 100 = 150%

KPI type	Manufacturing	Retail
Operational	Unplanned machine downtime (predictive maintenance); OEE	Stock-out rate; forecast accuracy
Quality	Defect rate found by AI camera inspection	Order accuracy; return rate
Financial	Maintenance cost; energy cost per unit; ROI	Sales uplift; average order value; inventory cost
Customer	On-time delivery to clients	CSAT / NPS; conversion rate from AI recommendations
Efficiency	Production cycle time	Chatbot resolution rate; checkout / response time

Manufacturing example – predictive maintenance

AI sensors listen to machine vibration and temperature and warn before a breakdown. KPIs tracked: ***unplanned downtime*** falls from 40 to 25 hours a month, ***maintenance cost*** falls, and ***OEE*** (overall equipment effectiveness) rises.

Retail example – AI recommendations & forecasting

An online store adds "customers also bought" recommendations and AI demand forecasting. KPIs tracked: ***conversion rate***, ***average order value***, ***stock-out rate*** and ***inventory holding cost*** – compared with the period before AI.

Mistakes to avoid when measuring AI impact

- ✗ No baseline – you cannot prove improvement without "before" numbers
- ✗ Measuring only model accuracy, not ***business results*** (money, time, customers)
- ✗ Ignoring hidden costs – data cleaning, training, maintenance
- ✗ Judging too early – AI benefits often grow over 6–12 months

Section C – Long Answer Questions

Q13 ***Discuss the major ethical issues in AI including bias, lack of transparency and privacy concerns. How do these affect business trust?***

AI ethics is about making sure AI is used in a way that is right, fair and safe for people and society. The main ethical issues are:

1. Bias and discrimination

AI learns from past data; if that data is unfair, the AI repeats and even increases the unfairness. Example: Amazon's hiring tool downgraded women's resumes (see Q5). Loan or insurance AI may reject people from certain areas unfairly.

2. Lack of transparency ("black box")

Many AI models (deep learning) cannot explain WHY they gave a decision. If a bank's AI rejects a loan and nobody can explain the reason, the customer feels cheated. ***Explainable AI (XAI)*** tries to solve this.

3. Privacy concerns

AI collects and analyses personal data – location, voice, face, shopping, health. Problems: data taken without consent, surveillance (facial recognition), data leaks, and using data for a different purpose.

4. Other issues

- **Accountability** – if a self-driving car or medical AI makes a mistake, who is responsible?
- **Misinformation & deepfakes** – fake videos / voices used for fraud and fake news
- **Job displacement** – automation replacing routine jobs
- **Manipulation** – AI nudging people to buy or believe things

How these affect business trust

- ✗ Customers stop using products they think are **unfair or spying** on them
- ✗ **Brand damage** – one viral news story of biased AI can hurt reputation for years
- ✗ **Legal penalties** under DPDP Act / GDPR and lawsuits
- ✗ **Employees resist** AI they do not understand or trust
- ✗ **Investors and partners** avoid companies with ethical risks

How businesses can reduce these ethical risks

- ✓ Use **diverse, balanced training data** and run regular bias audits
- ✓ Use **explainable AI** and give customers a reason for important decisions
- ✓ Collect data with **clear consent**; store it safely; delete it when no longer needed
- ✓ Keep **humans in the loop** for high-impact decisions (hiring, loans, medical)
- ✓ Create an **AI ethics policy / committee** and follow laws like the DPDP Act

Building trust – remember this formula

Trust = Fairness + Transparency + Privacy + Accountability. Companies build it with bias audits, explainable AI, clear consent, human review of important decisions (human-in-the-loop) and an AI ethics committee.

Q14 ***What are the key risks and challenges in AI adoption for businesses? Propose practical strategies for overcoming barriers related to data, talent, cost and stakeholder resistance. (merged: also "Discuss the challenges businesses face when adopting AI")***

AI adoption means bringing AI into everyday business processes. Many AI projects fail – not because the technology is bad, but because of the following challenges:

Challenge	What is the problem?	Practical strategy
1. Data	Poor quality, missing or scattered data (in different departments – "data silos")	Data governance, data cleaning, a central data warehouse / lake; start with data you already have
2. Talent	Shortage of data scientists and AI-skilled managers	Upskill / reskill staff, partner with universities & vendors, use low-code AI tools, hire "AI translators"

Challenge	What is the problem?	Practical strategy
3. Cost	High investment in technology, people and infrastructure; unclear ROI	Start with small pilot projects, cloud pay-as-you-go, ready-made AI services; track ROI
4. Stakeholder resistance	Employees fear job loss; managers trust their experience	Communicate clearly, involve staff early, training, show AI as an assistant, celebrate quick wins
5. Integration	Old (legacy) systems do not connect with new AI tools	APIs, phased upgrades, choose tools that fit existing software
6. Ethics & legal	Bias, privacy laws, unclear regulation	Responsible AI policy, audits, legal review
7. Security	Cyber attacks, data leaks, misuse of AI	Encryption, access control, regular testing
8. No clear strategy	AI used because it is trendy; projects stuck at pilot stage	Link every AI project to a business goal and a KPI

Real example – overcoming resistance

When UPS introduced its ORION routing AI (see Q17), many experienced drivers did not trust computer-made routes. UPS trained them, explained the savings, and let them give feedback and handle exceptions. Acceptance grew once drivers saw the routes worked.

Simple adoption roadmap

Identify a business problem -> check data -> small pilot -> measure KPIs -> train people -> scale up -> monitor & improve. Leadership support and change management are as important as the technology.

Q15 *Discuss how generative AI and intelligent automation are reshaping business models and organisational processes. What new roles and responsibilities are emerging for business professionals?*

Generative AI creates new content (text, images, code) – see Q8. ***Intelligent automation (IA)*** = ***RPA*** (software robots that copy rule-based tasks) ***+ AI*** (machine learning, language and vision), so that full processes – even those needing some judgement – run automatically.

Reshaping business models

- ***Hyper-personalisation at scale*** – a unique ad, email or product suggestion for every customer
- ***New AI-powered products & services*** – AI writing tools, AI tutors, AI design apps (subscription models)
- ***Lower cost of content & service*** – small firms can compete with big ones
- ***Faster innovation*** – product ideas, designs and prototypes in days instead of months
- ***24x7 digital service*** – AI agents serve customers without fixed office hours

Reshaping organisational processes

Department	How GenAI / intelligent automation changes the work
Marketing	Drafts ad copy, social media posts, product images; tests many versions quickly
Customer service	AI chatbots answer common queries; humans handle complex cases

Department	How GenAI / intelligent automation changes the work
Finance	Automatic invoice reading, payment matching, report summaries, fraud alerts
HR	Resume screening, job descriptions, onboarding chatbots, training content
IT / software	AI coding assistants (e.g. GitHub Copilot) write and test code faster
Legal / operations	Contract review, document summarising, process automation

Example: launching a marketing campaign

Step	Old way	With GenAI & automation
Market research	2-3 weeks of manual reading	AI summarises reviews & trends in hours
Ad content	One agency draft, slow changes	Dozens of versions in minutes, team picks best
Targeting	Same ad for everyone	Personalised message per customer group
Reporting	Manual Excel report weekly	Automatic dashboard + AI-written summary

New roles & responsibilities

New role	What they do
Prompt engineer / AI specialist	Designs prompts and workflows to get useful, accurate output from AI
AI product manager	Plans AI features that solve customer problems
AI ethics / governance officer	Ensures AI is fair, legal, safe and transparent
Analytics (AI) translator	Bridges business managers and data scientists
Human-in-the-loop reviewer	Checks and corrects AI output before it is used
Automation (RPA) developer	Builds software bots for repetitive processes

Responsibilities for every business professional: AI literacy, checking AI outputs (never blindly trusting them), protecting data, using AI ethically, managing teams of humans + AI, and continuous learning. AI changes jobs more than it removes them - people who use AI well will replace those who don't.

Q16 How can AI help improve customer satisfaction?

Customer satisfaction means how happy customers are with a product or service. AI improves it by making service faster, personal and available all the time:

1. **24x7 chatbots & virtual assistants** - instant answers any time (e.g. HDFC Bank's EVA chatbot)
2. **Personalised recommendations** - Amazon / Netflix suggest products and shows each person likes
3. **Faster delivery & accurate time estimates** - Swiggy / Zomato predict delivery time using AI
4. **Sentiment analysis** - AI reads reviews and social media to find unhappy customers quickly
5. **Predictive service** - spot problems before the customer complains (e.g. churn alerts, machine fault)

6. **Local language & voice support** – AI translates and speaks Hindi, Punjabi, Tamil, etc.
7. **Fraud protection** – AI blocks suspicious transactions, so customers feel safe
8. **Smart self-service** – track orders, returns and refunds without waiting in queues

Example – online food delivery

- AI recommends dishes based on past orders and time of day
- AI predicts delivery time and shows live tracking – fewer anxious calls
- Chatbot handles "where is my order?" and refunds instantly
- Result: faster service, fewer complaints, higher ratings and repeat orders

Remember

Measure with **CSAT, NPS and response time**. Always give customers an easy option to talk to a human – frustrating bots reduce satisfaction.

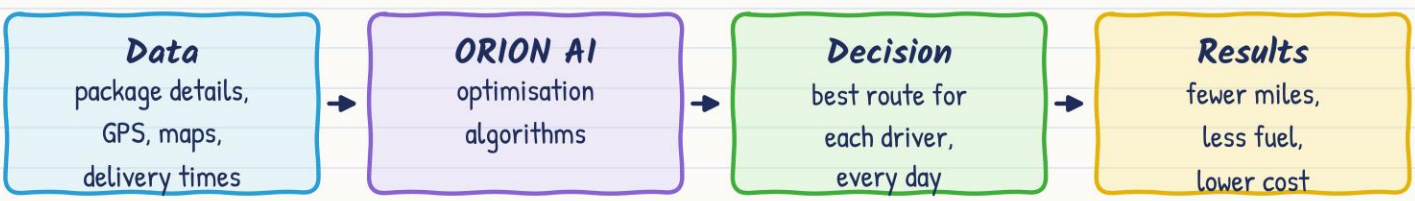
Section D – Case Study & Essay Questions

Q17 ***With the help of a real-life case study, analyse how a company has successfully implemented AI to transform its decision-making processes. Identify key success factors and lessons learned.***

Case study: UPS – ORION route optimisation

Company: UPS (United Parcel Service), one of the world's largest delivery companies. **ORION** = On-Road Integrated Optimization and Navigation.

Problem (before AI): Each driver delivers 100+ packages a day. The number of possible routes is astronomically large. Drivers planned routes using experience – fuel, time and money were wasted.



How AI transformed decisions: Route planning moved from the driver's judgement to an AI system that calculates the best route for every driver every morning, considering delivery time windows, traffic and package details. It took about 10 years to develop, and roll-out across US routes was planned to finish by the end of 2016.

Results (UPS estimates at full deployment, 2016, about 55,000 drivers): about **100 million fewer miles** and about **10 million gallons of fuel** saved per year, savings of around **US$300 – 400 million a year**, and about **100,000 tonnes less CO2**. ORION won the INFORMS Edelman Prize (2016), a top analytics award.

Key success factors

- **Clear business problem** with measurable value (miles, fuel, cost)
- **Strong data foundation** – years of package tracking and GPS data
- **Top management support** and long-term investment

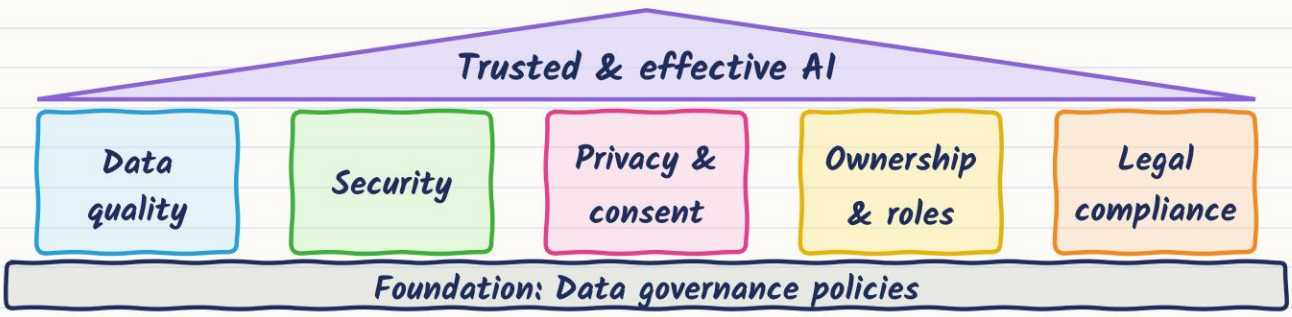
- ✅ **Involving drivers** – training, and drivers could give feedback and change routes when needed
- ✅ **Phased roll-out** – tested in a few centres first, then scaled
- ✅ **KPIs tracked** – miles per stop, fuel per route, on-time delivery

Lessons learned

- **People matter as much as technology** – drivers first resisted; change management made it work
- **Start with a high-value problem**, not with technology for its own sake
- **Human + AI works best** – AI suggests, experienced people handle exceptions
- **Data quality is the base** of any AI success
- **Keep improving** – the model is updated with new data (ORION later added real-time updates)

Q18 Explain the importance of data protection and data governance in AI-driven organisations.

Data protection = keeping personal and business data safe, private and legally used (security, consent, privacy). **Data governance** = the overall framework of rules, roles and processes to manage data quality, ownership and use (see Q9). Data protection is one important part of data governance.



Why they are important

1. **Accurate AI decisions** – well-governed, clean data gives reliable predictions; bad data gives wrong and biased results
2. **Legal compliance** – India's **DPDP Act 2023** (penalties up to Rs 250 crore) and Europe's **GDPR** (up to 4% of global turnover) require consent, purpose limitation and safety of personal data
3. **Customer trust** – people share data only with companies they trust
4. **Security** – protects against hacking, data leaks and insider misuse
5. **Fairness & ethics** – checking data for bias before training AI
6. **Accountability** – clear data owners; AI decisions can be audited and explained
7. **Efficiency** – no duplicate or conflicting data; teams find the right data quickly

What goes wrong WITHOUT them

- ❌ Biased or wrong AI decisions from dirty, incomplete data
- ❌ Data breaches – heavy fines, lawsuits and news headlines
- ❌ Nobody knows who owns or approved the data – no accountability
- ❌ Customers withdraw consent and move to competitors

Good practices

- Take clear ***consent*** and collect only the data needed (***data minimisation***)
- ***Encrypt*** data, use access control and strong passwords
- ***Anonymise / mask*** personal data before using it for AI training
- Appoint ***data owners / stewards*** and a data protection officer
- Data catalog, retention & deletion policy, and ***regular audits***

Q19 Explain the legal, ethical and social implications of AI adoption.

When a company adopts AI, the effects go beyond the company – they touch the law, moral values and society as a whole.

A. Legal implications

- ***Data protection laws*** – DPDP Act 2023 (India), GDPR (Europe): consent, data rights, heavy fines
- ***AI regulation*** – the EU AI Act (2024) classifies AI by risk: unacceptable (banned), high, limited and minimal risk; high-risk AI (hiring, credit, health) has strict rules
- ***Liability*** – who pays if AI causes harm: the company, the developer or the user?
- ***Intellectual property*** – who owns AI-generated content? Was copyrighted data used for training?
- ***Consumer & employment law*** – no unfair or discriminatory decisions in hiring, lending or pricing

India focus – DPDP Act 2023 in simple words

- Companies ("***data fiduciaries***") must take clear ***consent*** and use data only for the stated purpose
- People ("***data principals***") can access, correct and erase their data
- Companies must keep data safe and report breaches; the ***Data Protection Board*** can impose penalties

B. Ethical implications

- ***Bias & fairness*** – AI must treat all groups equally
- ***Transparency*** – people should know when AI is used and why it decided something
- ***Privacy*** – respect personal data; no hidden surveillance
- ***Accountability*** – humans stay responsible for AI decisions
- ***Honesty*** – no deepfakes, fake reviews or manipulative AI marketing

C. Social implications

Positive impact	Negative impact / concern
Better healthcare, education and banking access, even in villages	Job loss in routine work; need for reskilling
Higher productivity & economic growth	Digital divide – those without skills or internet fall behind
Help for people with disabilities (voice, translation)	Misinformation, deepfakes and loss of trust
Faster, safer services (fraud detection, disaster alerts)	Over-dependence on AI; less human interaction

Apply it: one example, three lenses – an AI hiring tool

- **Legal:** must follow data protection and anti-discrimination laws; candidates' data needs consent
- **Ethical:** must not reject candidates because of gender, caste or age; decisions should be explainable
- **Social:** can speed up hiring for millions, but if biased it can shut groups out of jobs and widen inequality

How should businesses respond?

- Set up an **AI governance policy** and an ethics / risk committee
- Do **legal checks** before launching AI (privacy, IP, sector rules)
- **Reskill employees** whose jobs change; create new roles
- Be **transparent** with customers – tell them when AI is used
- Use AI for **social good** too – accessibility, local languages, rural services

Conclusion (write this at the end)

AI brings big benefits, but only if it is adopted **responsibly** – following the law, respecting ethics and caring for society. Companies should combine AI governance, reskilling of employees and transparency with customers, so that AI creates value for business AND society.

Answer Plans for 10-Mark Questions

Use these skeletons to write long answers in the exam – each heading becomes a paragraph or a set of points.

Q	Headings to write (in order)
Q10	Intro: business analytics -> ladder diagram -> comparison table (3 industries) -> diagnostic (extra) -> how they support AI decisions -> airline example -> conclusion
Q13	Intro: AI ethics -> bias -> transparency -> privacy -> other issues -> effect on trust -> solutions -> trust formula
Q14	Intro -> challenge table (data, talent, cost, resistance + 4 more) -> strategies -> UPS example -> adoption roadmap -> conclusion
Q15	Define GenAI & intelligent automation -> new business models -> process changes (table) -> campaign example -> new roles -> responsibilities
Q17	Company & problem -> ORION diagram -> how decisions changed -> results (numbers) -> success factors -> lessons learned
Q18	Define data protection & governance -> pillars diagram -> 7 reasons -> what goes wrong -> good practices -> conclusion
Q19	Intro -> legal (DPDP, EU AI Act, liability, IP) -> ethical -> social (table) -> hiring-tool example -> business response -> conclusion

Last-Minute Revision - One-Line Answers

Term	Remember in one line
Descriptive analytics	Past data -> "What happened?" (dashboards, reports)
Diagnostic analytics	"Why did it happen?" (drill-down, root cause)
Predictive analytics	Forecast -> "What will happen?" (ML, regression)
Prescriptive analytics	Best action -> "What should we do?" (optimisation, AI)
KPI	Measurable value that tracks progress towards a key goal (SMART)
AI KPIs	ROI, cost saved, time saved, forecast accuracy, CSAT / NPS, downtime
AI bias	Unfair AI results due to biased data - e.g. Amazon hiring tool
Responsible AI	Fair, transparent, accountable, private, safe, inclusive AI
Data privacy	Person's right to control their personal data - consent is key
Data governance	Rules, roles & processes for accurate, secure, proper data use
Data protection	Keeping personal data safe & legal (DPDP Act 2023, GDPR)
Generative AI	Creates new text, images, code from prompts - ChatGPT, Claude, Gemini
Intelligent automation	RPA (software bots) + AI = end-to-end automated processes
Black box / XAI	AI that can't explain decisions / Explainable AI fixes this
Hallucination	GenAI confidently giving wrong or made-up information
Human-in-the-loop	A person reviews AI decisions before final action
AI adoption barriers	Data, talent, cost, resistance, integration, ethics, security, strategy
Case studies	Netflix (recommendations), UPS ORION (routes), Amazon (hiring bias)
EU AI Act (2024)	Risk-based AI law: unacceptable, high, limited, minimal risk

Short form	Full form	Short form	Full form
AI / ML	Artificial Intelligence / Machine Learning	KPI	Key Performance Indicator
ROI	Return on Investment	CSAT	Customer Satisfaction Score
NPS	Net Promoter Score	OEE	Overall Equipment Effectiveness
RPA	Robotic Process Automation	IA	Intelligent Automation
LLM	Large Language Model	NLP	Natural Language Processing
XAI	Explainable AI	GenAI	Generative AI
DPDP	Digital Personal Data Protection Act, 2023 (India)	GDPR	General Data Protection Regulation (EU)

Golden line for any AI essay

"AI does not replace good managers - it gives them better information. The best results come when ***data, technology and people*** work together ***responsibly***."